

BRIEF REPORT

Paradoxical Sleep Deprivation Retards Extinction of Conditioned Taste Aversion

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Conditioned taste aversion (CTA) induced by 15-min saccharin drinking followed 30 min later by poisoning (0.15 M lithium chloride, 2% body weight) was extinguished by repeated exposure of water-deprived hooded rats to saccharin (15-min sessions at 3- to 4-day intervals). Animals subjected to paradoxical sleep deprivation (PSD) for 24 hr before each extinction session attained full CTA extinction in six sessions, normally sleeping animals in two sessions. It is concluded that PSD interferes with the modification of CTA engrams.

The rapidly growing research into the effects of preacquisition or post-acquisition paradoxical sleep deprivation (PSD) on learning (for review, see Fishbein & Gutwein, 1977; Hennevin & Leconte, 1977) lead to formulation of the hypothesis (Greenberg & Pearlman, 1974) that PSD disrupts slow learning of difficult tasks more than the acquisition of rapidly established, "prepared" associations. Contrary to the above hypothesis, 24-hr preacquisition PSD has recently been reported (Danguir & Nicolaidis, 1976) to prevent learning of the conditioned taste aversion (CTA). We have confirmed this finding (Venkatakrishna-Bhatt, Bureš, & Burešová, 1978) and shown that 24-hr PSD which blocks CTA acquisition does not interfere with retrieval of already established CTA engrams. These results suggested that PSD impairs the formation of the short-term gustatory trace and/or the conversion of this trace to the long-term CTA engram, but that it does not prevent recognition of the previously acquired CTA. The purpose of the present experiment is to investigate the effect of PSD on CTA extinction, that is, on the retrieval-induced modification of CTA engram.

The experiments were performed in 28 male hooded rats aged 2 to 3 months. The animals were maintained on a 23.75-hr water deprivation schedule, with water available for only 15 min daily, in a testing apparatus consisting of a plastic enclosure (48 × 15 × 30 cm) with a calibrated pipet attached to the narrow front wall 6 cm above the floor level. The liquids

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consumed were measured with 0.1-ml accuracy. Water-deprived rats were given 15-min access to water on Days 1 and 2. On Day 3 water was substituted by 0.1% sodium saccharin, followed 30 min later by intraperitoneal injection of 0.15 *M* lithium chloride (2% body weight), and within a few minutes elicited severe but fully reversible symptoms of gastrointestinal distress, serving as a standard US in the CTA paradigm. During the subsequent 22 days experimental rats had daily access (15 min) to a drinking spout filled either with water or (on every third or fourth day) with saccharin solution. If, due to strong CTA, spontaneous saccharin intake dropped below 8 ml, the animals were restrained and received additional saccharin by forced drinking, i.e., by slow intraoral infusion, so that each rat consumed at least 8 ml of saccharin on each of the six extinction trials. While the first experimental group (E_1 , $n = 10$) received no further treatment, the second group (E_2 , $n = 10$) was exposed to 24-hr PSD before each extinction trial. The control group (C , $n = 8$) was given water throughout the experiment, except on Day 3, when it was subjected to standard CTA training.

Paradoxical sleep deprivation was induced by a modified pedestal technique (Morden, Mitchell, & Dement, 1967): The pedestals (7 cm in diameter and 7 cm high) were placed in $20 \times 20 \times 40$ -cm enclosures with electrifiable grid floors. As many as 20 animals could be simultaneously treated in such compartments connected to the same shock source (2 mA, 50 Hz, 1 sec on, 3 sec off). Rats were adapted to the platform by being placed onto 80-cm-high pillars with circular upper surfaces (7 cm in diameter) for 2 to 3 hr before the first PSD. This technique prevents paradoxical sleep in the same way as the standard "water tank" method which cannot be employed in CTA experiments since the availability of water precludes reliable water deprivation. After each PSD treatment the animals were offered food in their home cages for 30 min before being placed into the drinking box. Differences in fluid intake during acquisition and extinction were examined using Student's *t* test for correlated values. One-way analysis of variance was used for between-group comparisons of fluid intake on various days of the experiment.

Figure 1 shows saccharin (experimental groups) or water (control group) consumption during 15-min testing periods, starting with the learning session on Day 3, when all three groups received saccharin solution as the CS. Between-group differences in saccharin consumption were not statistically significant [$F(2, 25) = 2.4$, $p > 0.05$]. The first extinction trial on Day 7 revealed well-developed CTA in both experimental groups, the volume of spontaneously consumed saccharin decreasing to 1.5 and 0.8 ml in groups E_1 and E_2 , respectively. Corresponding water intake in the control group was not significantly lower than saccharin consumption before CTA acquisition [Day 3, $t(9) = 0.3$, not significant]. The stable level of water consumption in the control group indicated that the

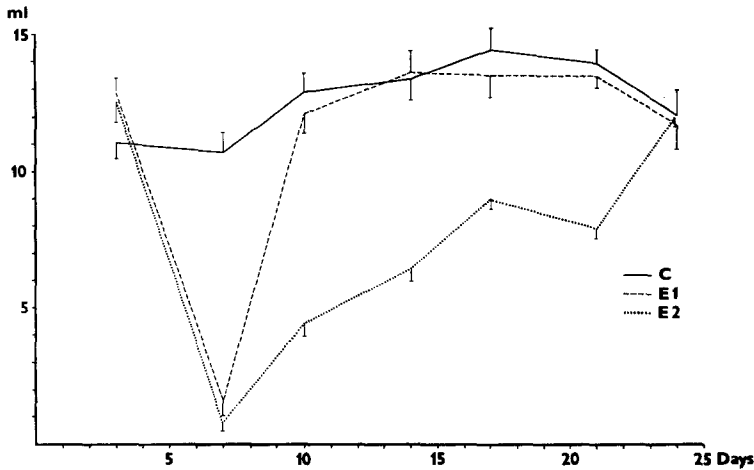


FIG. 1. Extinction of the conditioned taste aversion to saccharin. Ordinate: volume of water (Group C) or saccharin (Groups E₁ and E₂) intake. Abscissa: days of experiment. The vertical bars denote SEM values. All groups received saccharin followed by lithium chloride injection during acquisition on Day 3. Group C was then offered water, and Groups E₁ and E₂ were offered saccharin during extinction trials. The extinction sessions were preceded by 24-hr PSD in Group E₂.

nonspecific effects of CTA treatment were negligible. Extinction was rapid in experimental group E₁, where two trials were sufficient to restore the normal saccharin intake. On the contrary, experimental group E₂ with PSD preceding all extinction trials extinguished slowly and still showed significant aversion in the fifth extinction trial [$F(2, 25) = 11.4, p < 0.01$]. Only the sixth extinction trial restored the saccharin intake to the preaversion level.

While most PSD studies concentrate on the effects of preacquisition and/or postacquisition PSD on learning, little is known about the PSD effects on extinction. If extinction is viewed as negative learning, it is not surprising that it is impaired by the pretrial PSD similarly as the acquisition of the same task. In fact, Deweer (1970) reported that experimental modifications of vigilance affect acquisition and extinction in a similar way, probably by influencing the mechanism of consolidation, common to both processes. This reasoning is perhaps not directly applicable to CTA, however, which seems to have different mechanisms of acquisition and extinction. As shown by Burešová and Bureš (1975) forced feeding saccharin to rats under bilateral cortical spreading depression (CSD) is an inadequate CS for CTA acquisition in naive rats, but can induce extinction of CTA acquired earlier. Intraperitoneal injection of 2% saccharin followed by lithium chloride intoxication elicits conditioned saccharin aversion in intact rats but not in rats anesthetized with pentobarbital (Burešová & Bureš, 1977). On the other hand, the same intraperitoneal

injection of saccharin into CTA-trained animals induces extinction in both intact and anesthetized rats. The differential effect of CSD and anesthesia on CTA acquisition and extinction suggests that these interventions do not interfere with the neural processes fitting the CTA engram with a neutral label. This is in good agreement with the fact that formation of the permanent CTA engram is not prevented by CSD (Burešová & Bureš, 1973) or anesthesia (Burešová & Bureš, 1977; Millner & Palfai, 1975; Roll & Smith, 1972), applied after administration of the gustatory CS but before poisoning. On the other hand, impairment of both CTA acquisition and extinction may indicate that the PSD is a strong treatment which blocks the common link of both processes, perhaps of the formation or modification of long-term engrams (Fishbein, 1970; Fishbein & Gutwein, 1977; Sagales & Domino, 1973). Such a view is consistent with the finding that PSD does not impair but rather improves CTA retrieval (Venkatakrishna-Bhatt *et al.*, 1978), that is, a process which does not require any modification of the CTA trace.

Finally, the possibility must be considered that the difference between groups E_1 and E_2 is due not only to the lack of paradoxical sleep but also to nonspecific effects of the PSD procedure (stress, confinement, fatigue). Although these factors do not contribute significantly to PSD-induced interference with the acquisition of other tasks (see review by Fishbein & Gutwein, 1977), they may be important in the case of learning involving visceral signals. Further analysis of the experimental procedures differentially influencing CTA acquisition, retention, and extinction should contribute to a better understanding of the mechanism of CTA.

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